

A note on the NP Hardness of the maximum profit concurrent open shop problem

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Consider the following variation of the open shop scheduling problem. A set of n jobs should be processed on m machines. Each job i consists of m tasks, one on each machine. The processing time of job i on machine j is denoted by p_{ij} and assumed to be integer. Tasks of the same job can be processed on different machines concurrently. The revenue obtained from job i is $w_i \alpha^{C_i}$. Where w_i is the weight of the job, C_i is the completion time of the last task of job i and $0 < \alpha < 1$ is a discount factor. The problem was first introduced by Bukchin and Gerchak (2014) in the context of decentralized scheduling. Other variations of the problem, without profit maximization, were studied by Wagner and Sriskandarajah (1993), Ng et al. (2003), Leung et al. (2005a), and Leung et al. (2005b). Roemer (2006) proved the NP-Hardness of some variants of the problem.

Theoretically, since α is assumed to be any real number in the interval $(0,1)$ it is impossible to calculate the value of the objective function in finite number of steps. In practice, it is typically assumed that real numbers can be approximated sufficiently well by floating point data structures and that all the basic arithmetic operations on floating points can be carried out in constant time. Only under such assumptions it makes sense to discuss the computational complexity of the problem in a discrete computation model.

Claim: The above problem is NP-Hard in the ordinary sense even for $m = 2$ and under the assumption that basic arithmetic operations can be applied in constant time.

Proof: We prove this claim by reduction from the subset sum problem, see Garey and Johnson (1979), page 223. Recall that the subset sum is a decision problem with input $a_1, \dots, a_n, b$ (all integers) where one need to determine whether or not a subset $S \subset \{1, \dots, n\}$ exists such that $\sum_{i \in S} a_i = b$.

Given an instance of the subset sum problem we construct an instance of our problem with $n + 1$ jobs and $m = 2$ machines. We set $p_{i1} = a_i$ for all $i = 1, \dots, n$, $p_{n+1,1} = M$, $p_{i2} = 0$, $p_{n+1,2} = M + b$ where M is a sufficiently large number; $w_i = a_i$ for all $i = 1, \dots, n$ and w_{n+1} is calculated as described below.

The idea of the reduction is to set the values of the remaining parameters (α, w_{n+1}, M) such that in any optimal solution:

1. job $n + 1$ is scheduled to be completed at the earliest possible time, which is $M + b$.
2. If a subset $S \subset \{1, \dots, n\}$ exists such that $\sum_{i \in S} a_i = \sum_{i \in S} p_{i1} = b$ then these jobs are scheduled on machine 1 during the interval $[0, b]$

That is if and only if the answer to the subset sum decision problem is “yes” the optimal solution of maximum profit concurrent open shop problem satisfy the second condition.

For the first condition to hold the parameters must satisfy

$$w_{n+1} (\alpha^{M+b} - \alpha^{M+b+1}) \geq \sum_{i=1}^n a_i \quad (1)$$

The left hand side of this inequality represents the loss caused by postponing the completion time of job $n + 1$ by one time unit compared with its minimal possible completion time. The right hand side of the inequality represents a loose upper bound on the gain that can be obtained from jobs $1, \dots, n$ in any schedule. Clearly if the loss from postponing job $n + 1$ by one unit is greater than the total profit that can be gained from all the other jobs in any schedule, then in any optimal solution job $n + 1$ must be completed at the earliest possible time. Clearly, for any $0 < \alpha < 1$ and M it is possible to select w_{n+1} to be large enough such that the inequality is satisfied. For example

$$w_{n+1} = \left\lceil \frac{\sum_{i=1}^n a_i}{\alpha^{M+b} - \alpha^{M+b+1}} \right\rceil$$

Next, since in an optimal solution (assuming inequality (1) holds) job $n + 1$ is completed at time $M + b$ its processing on machine 1 can start at any time in the interval $[0, b]$ and we can assume without loss of generality that it is scheduled to start at time b . Next we are looking for conditions on the parameters that will guaranty keeping machine 1 busy processing jobs from $\{1, \dots, n\}$ throughout the interval $[0, b]$ if this is possible. I.e., if there is a subset $S \subseteq \{1, \dots, n\}$ such that $\sum_{i \in S} a_i = b$. Note that under such a schedule the gain from jobs $\{1, \dots, n\}$ is bounded below by $b\alpha^b$ because the smallest possible gain from a job scheduled in this interval is $w_i \alpha^b = a_i \alpha^b$; additional gain will also be obtained from jobs that will be processed after time $b + M$. On the other hand if machine 1 does not uses the whole interval $[0, b]$ for jobs from the set $\{1, \dots, n\}$, then a upper bound on the possible revenue from the jobs of this set is $(b - 1) + \sum_{i=1}^n a_i \alpha^{M+1}$. Hence condition 2 must hold if

$$b\alpha^b \geq (b - 1) + \sum_{i=1}^n a_i \alpha^{M+1} \quad (2)$$

To finalize the proof we need to find α, M that satisfy (2). Note that by selecting M we can make the term $\sum_{i=1}^n a_i \alpha^{M+1}$ to be as close to zero as we desire so it is enough to find a value of α such that $b\alpha^b = (b - 1) + \frac{1}{2}$.

By solving this equation for α we obtain

$$\alpha = \sqrt[b]{\frac{b - 1/2}{b}}$$

Now we can set M to satisfy

$$\sum_{i=1}^n a_i \alpha^{M+1} \leq \frac{1}{2}$$

i.e.,

$$\alpha^{M+1} \leq \frac{1}{2 \sum_{i=1}^n a_i}$$

$$\log \alpha^{M+1} \leq \log \frac{1}{2 \sum_{i=1}^n a_i}$$

$$M \geq \frac{-\log(2 \sum_{i=1}^n a_i)}{\log \alpha}$$

(we switched the direction of the inequality because $\log \alpha < 0$). Now since the processing times are integer we set

$$M = \left\lceil \frac{-\log 2 \sum_{i=1}^n a_i}{\log \alpha} \right\rceil$$

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Example: consider an instance of “subset sum” with $b = (2,4,3)$ $b = 5$. The answer to this decision problem is obviously “Yes” since $2+3=5$.

The problem can be reduced to an instance of the maximum profit concurrent open shop problem as follow.

- $n = 4$
- $b = 5$
- $p_{11} = 2, p_{12} = 4, p_{13} = 3$
- $p_{21} = p_{22} = p_{23} = 0$
- $\alpha = \sqrt[b]{\frac{b-1/2}{b}} = \sqrt[5]{\frac{5-1/2}{5}} = \sqrt[5]{\frac{9}{10}} = 0.9^{0.2}$
- $M = \left\lceil \frac{-\log 2 \sum_{i=1}^n a_i}{\log \alpha} \right\rceil = \left\lceil \frac{-\log 18}{0.2 \log 0.9} \right\rceil = \left\lceil \frac{5 \log \frac{1}{10}}{\log \frac{9}{10}} \right\rceil = 110$
- That is $p_{14} = 110, p_{24} = 115$
- Now the value of w_{n+1} should satisfy $w_{n+1} (\alpha^{M+b} - \alpha^{M+b+1}) \geq \sum_{i=1}^n a_i$. We select

$$w_{n+1} = \frac{\sum_{i=1}^n a_i}{\alpha^{M+b} - \alpha^{M+b+1}} = \frac{9}{(0.9^{0.2})^{115} - (0.9^{0.2})^{116}} = \frac{9}{0.9^{23} - 0.9^{23.2}} \approx 4869.95$$

$p_{14} =$

It is easy to check that in any optimal solution jobs 1 and 3 are scheduled first on machine 1 during the interval $[0,5]$, job 4 in the interval $[5,115]$ and then job 2. On machine 2, job 4 is scheduled in the interval $[0,115]$. The value of the optimal solution is

$$2 \times 0.9^{0.4} + 3 \times 0.9^1 + 4869.95 \times 0.9^{23} + 4 \times 0.9^{23.8} \approx 436.56$$

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